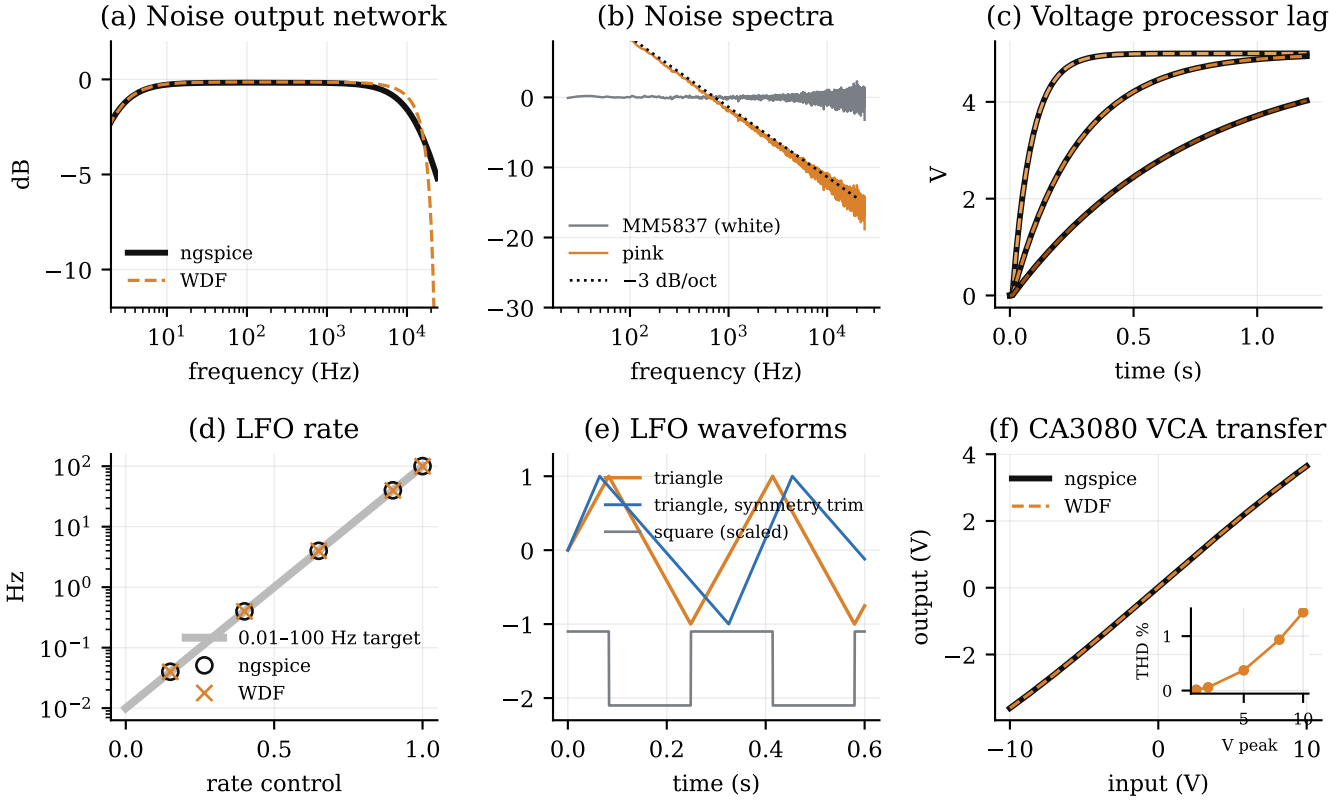
(a) Noise output network
dB
0
−5
−10
ngspice
WDF
10^1
10^2
10^3
10^4
frequency (Hz)
(b) Noise spectra
0
−10
−20
−30
MM5837 (white)
pink
−3 dB/oct
10^2
10^3
10^4
frequency (Hz)
(c) Voltage processor lag
V
4
2
0
0.0
0.5
1.0
time (s)
(d) LFO rate
Hz
10^2
10^1
10^0
10^{-1}
10^{-2}
0.01–100 Hz target
ngspice
WDF
0.0
0.5
1.0
rate control
(e) LFO waveforms
1
0
−1
−2
triangle
triangle, symmetry trim
square (scaled)
0.0
0.2
0.4
0.6
time (s)
(f) CA3080 VCA transfer
output (V)
4
2
0
−2
ngspice
WDF
THD %
1
0
5
10
V peak
−10
0
10
input (V)